

Q1)

a) We define our page replacement algorithm:

- i) When a new page is admitted into the physical memory, its corresponding counter is set to 1
- ii) When the string reference does not produce a page fault, we then increment the by 1 counter
- iii) When the string reference does produce a page fault, we then decrement the counter by 1. Once the counter goes below 0, it is now time to put the new frame with a counter set to 1
- iv) We keep track every time the value of the counter is incremented or decremented, or even its corresponding frame has changed, in a LRU fashion.

b) Steps of the algorithm in placed:

Reference String: 1,2,3,4,5,3,4,1,6,7,8,7,8,9,7,8,9,5,4,5,4,2

the 1), 2), 3), ... represent the step number. The numbers in blue represents the value of counter corresponding to each frame.

1)			2)			3)			4)			5)			6)			7)			8)			9)		
1	1		0	1		0	1		0	1		0	1		1	5		1	5		1	5		1	5	
1	2		1	2		0	2		0	2		0	2		0	2		0	2		0	2		1	1	
1	3		1	3		1	3		0	3		0	3		0	3		1	3		1	3		1	3	
1	4		1	4		1	4		1	4		0	4		0	4		0	4		1	4		1	4	
10)			11)			12)			13)			14)			15)			16)			18)			19)		
0	5		0	5		0	5		0	5		1	6		1	6		1	6		1	6		1	6	
1	1		1	1		1	1		0	1		0	1		0	1		0	1		0	1		0	1	
1	3		0	3		0	3		0	3		0	3		1	7		1	7		2	7		2	7	
1	4		1	4		0	4		0	4		0	4		0	4		1	8		1	8		2	8	
20)			21)			22)			23)			24)			25)			26)			27)			28)		
1	6		1	6		1	6		1	6		0	6		0	6		0	6		0	6		1	5	
1	9		1	9		1	9		2	9		2	9		2	9		2	9		1	9		1	9	
2	7		3	7		3	7		3	7		3	7		2	7		2	7		2	7		2	7	
2	8		2	8		3	8		3	8		3	8		3	8		2	8		2	8		2	8	
29)			30)			31)			32)			33)			34)			35)			36)			37)		
1	5		1	5		0	5		1	4		1	4		2	4		2	4		2	4		2	4	
1	9		1	9		0	9		0	9		1	5		1	5		1	5		1	5		0	5	
1	7		1	7		1	7		1	7		1	7		1	7		0	7		0	7		0	7	
2	8		1	8		1	8		1	8		1	8		1	8		1	8		0	8		0	8	
38)			39)																							
1	4		1	4																						
0	5		0	5																						
0	7		1	2																						
0	8		0	8																						

of page fault: 13

no page fault: 9

c) What is the minimum number of page faults for an optimal page replacement strategy for the reference string in part b with four page frames?

	1	2	3	4	5	3	4	1	6	7	8	7	8	9	7	8	9	5	4	5	4	2
1)																						
	1		1		1		7		7		7		7		2							
	2		5		5		5		5		5		5		5							
	3		3		6		6		8		8		8		8							
	4		4		4		4		4		9		4		4							

of page fault: 11

no page fault: 11